Data Analysis for Mechanical System Identification
Final Project

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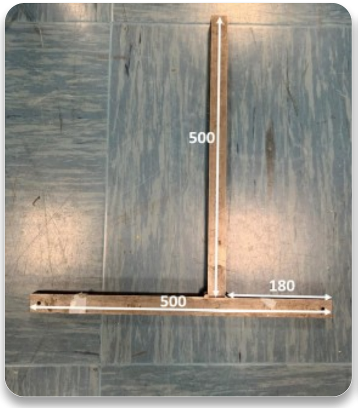
Contents

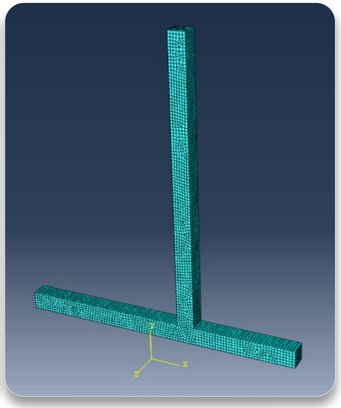
Project Description and Goals

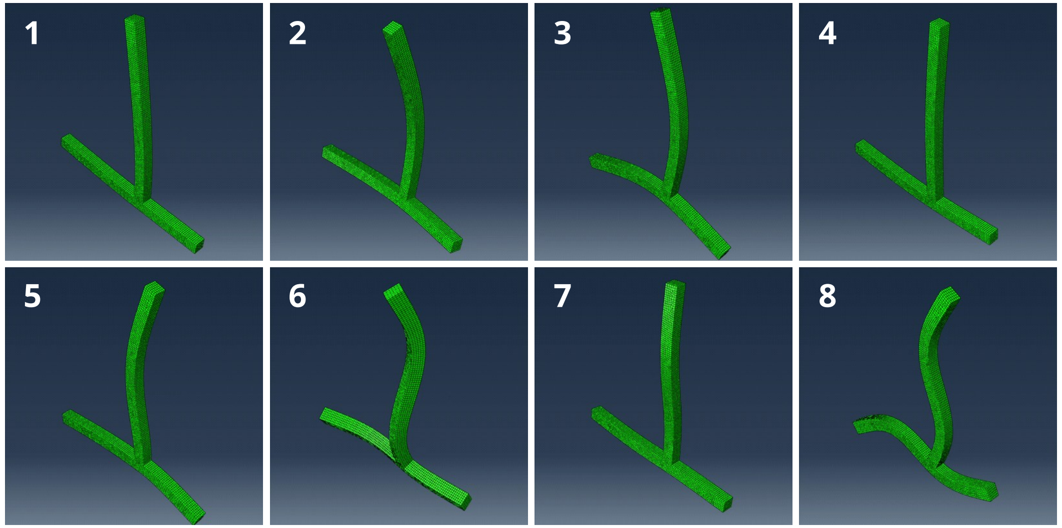
- The objective of the project is to perform a **modal analysis** of a T-shaped steel specimen by extracting the modal parameters via an impact hammer test and reconstructing its FRF.
- The mass of the specimen is 7,007 grams.
- The cross-sectional area of the specimen is 30x30 mm^2.
- The specimen is held in **free-free** conditions, which is a boundary condition where the structure is not constrained or fixed at any end and is “free” to translate and rotate in space (6 degrees of freedom).

FEM Model for Mode Shapes and Eigenfrequencies Identification

- The FE model was designed in Abaqus, the geometry, properties and boundary conditions were matched to the given structure.
- The part was meshed with 5mm elements, and **8 modes** of interest were obtained (seen in animations below). The first 6 modes are neglected.







Eigenfrequencies (Hz)	
1	178.68
2	402.36
3	485.17
4	605.61
5	654.59
6	1233.90
7	1341.30
8	1383.30

Steel Properties	
Density ρ	7,786 kg/m³
Young's Modulus E	200 GPa
Poisson's Ratio ν	0.3

Identifying the Optimal Sensor Configuration Using AutoMAC

- Initial sensor positions were chosen in regions with **large displacements**, to ensure strong dynamic responses, however, the **key goal** was to record **distinct modal behaviors**, minimizing spatial correlation between modes.
- The **AutoMAC** algorithm was the key tool to validate each sensor configuration.

$$AutoMAC\left(A_i, A_j\right) = \frac{\left|\left\{\psi_{A,i}\right\}^T \left\{\psi_{A,j}\right\}\right|^2}{\left(\left\{\psi_{A,i}\right\}^T \left\{\psi_{A,i}\right\}\right) \cdot \left(\left\{\psi_{A,j}\right\}^T \left\{\psi_{A,j}\right\}\right)}$$

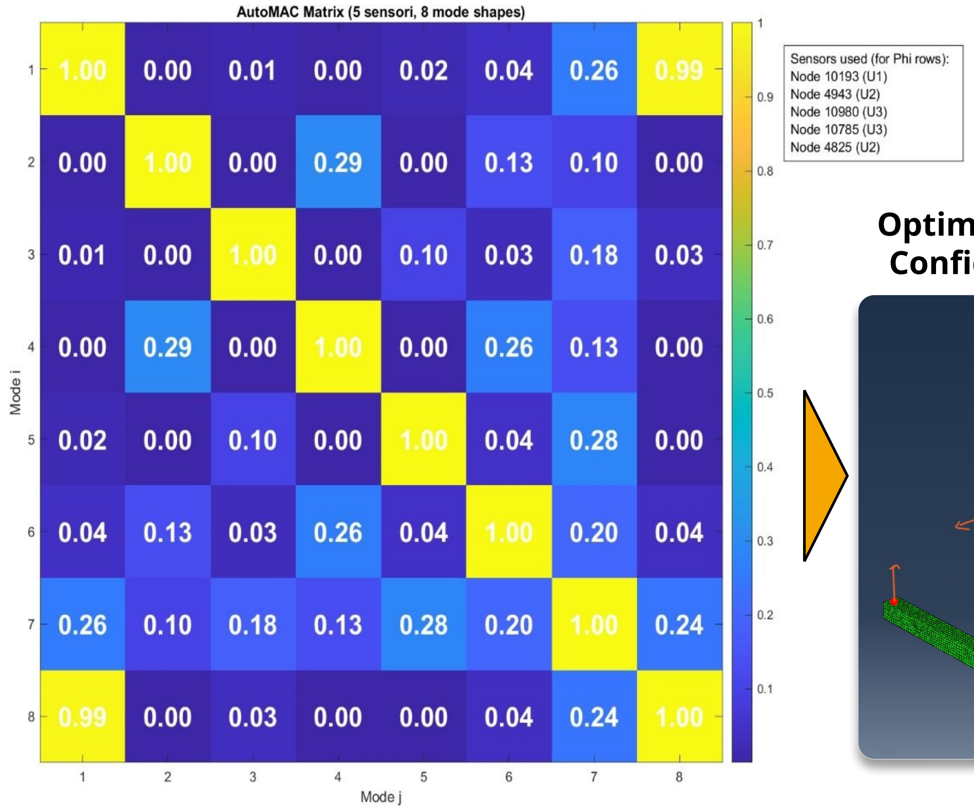
$\psi_{A,i}$ and $\psi_{A,j}$ are the i^{th} and j^{th} eigenmodes of the structure A

- The AutoMAC **matrix** compares mode shapes across sensor locations to assess spatial correlation, showing **how much each sensor contributes to each mode.**

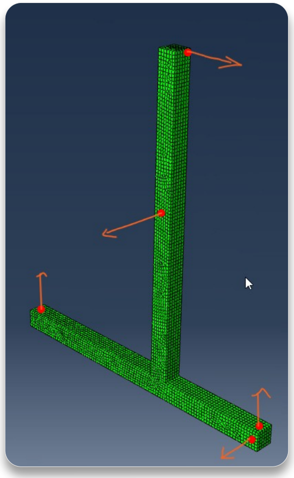
Optimal Sensor Configuration

- After multiple iterations:
- We selected a configuration where most off-diagonal values showed weak or no correlation, confirming good mode separation.
- **Mode 1 and mode 8** showed high correlation — acceptable since their frequencies are **far apart** and won't interfere during identification.
- There is no risk of misinterpreting the modes, and the selected configuration enables **efficient use** of the 5 accelerometers.

Final AutoMAC Matrix



Optimal Sensor Configuration



Experiment in the Lab
Structure & Sensor Mounting

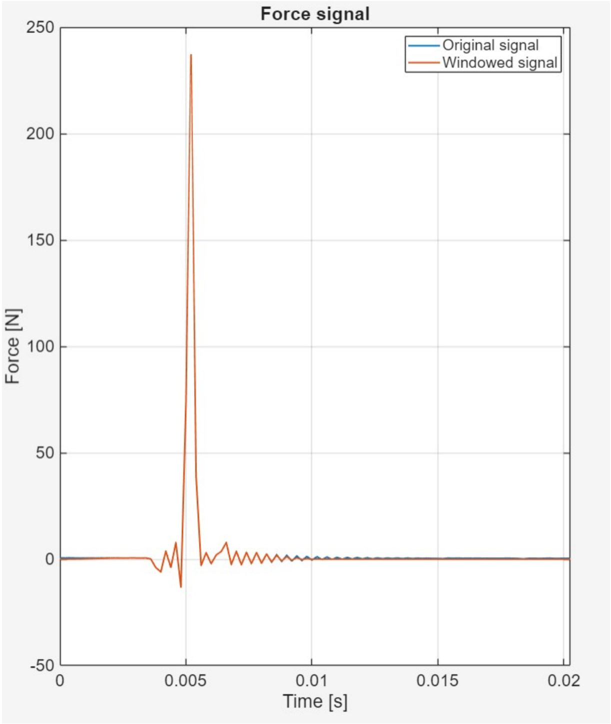
- The structure was suspended using nylon cords to achieve approximate **free-free** boundary conditions.
- 5 accelerometers were mounted, considering their measurement direction.
- Small **plastic intermediate parts** (acting as low pass filters) and glue were used for mounting.
- Proper installation is essential to prevent noise from being introduced at the input.

Experiment in the Lab
Hammer Testing & Impulse Signal

- An **impact hammer** was used to apply an **impulse** excitation, and a load cell is integrated to measure the force input.

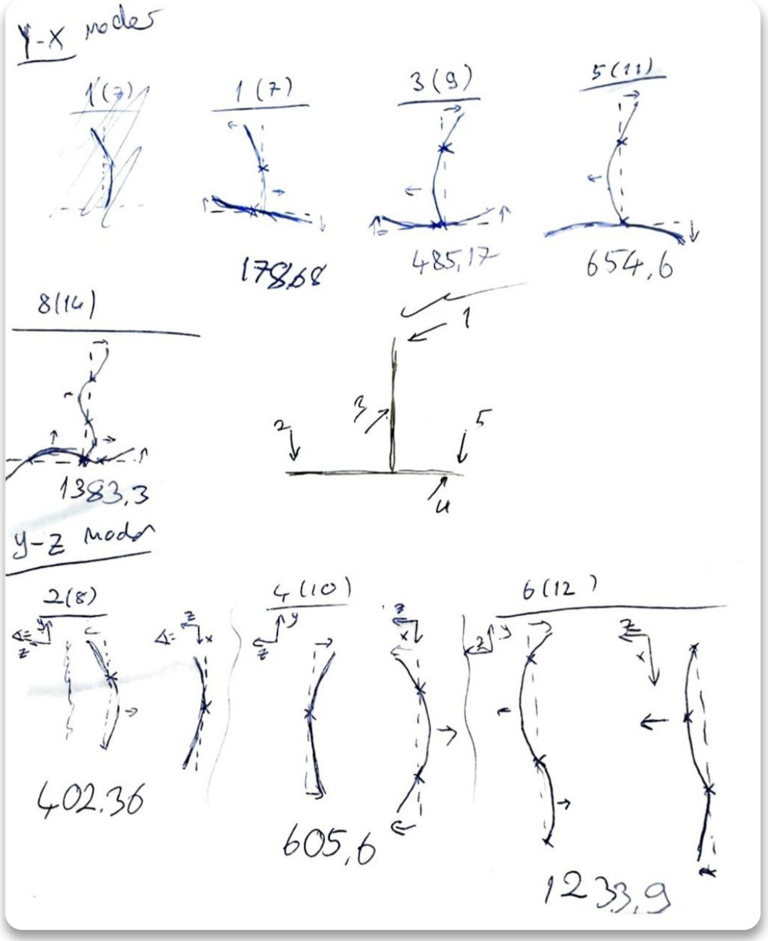
- Regarding the **impulse signal**:
  - It is cost-effective, fast and easy to apply
  - Excites a wide frequency range.
  - No load effect.
  - Operator-dependent process (variations in strength and location of strength, potential for double hits)

- Among others, plastic tip was selected to observe results in the frequency-range of interest.



Experiment in the Lab
Impact Locations

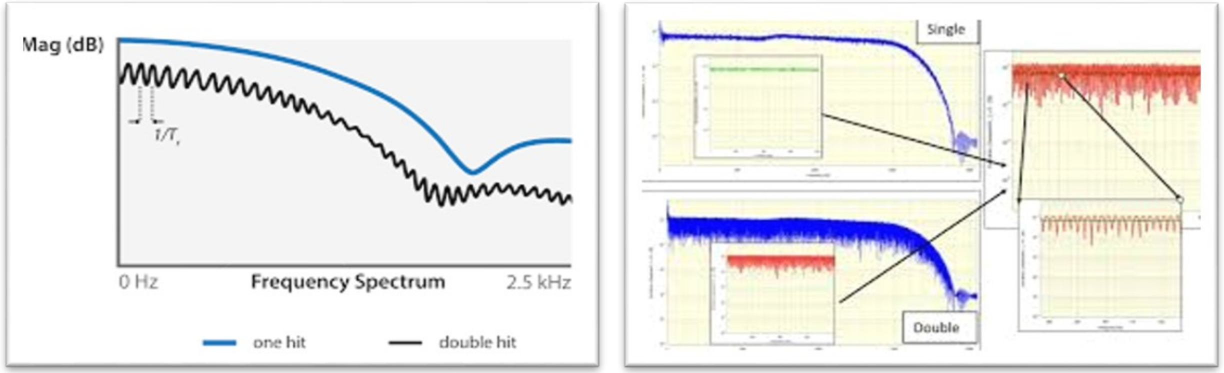
- 4 impact points were selected to properly excite all 8 modes.
- It was crucial that at least one of these was **collocated** (impact point being the same position of a sensor)
- Direct and accurate estimation of the FRF



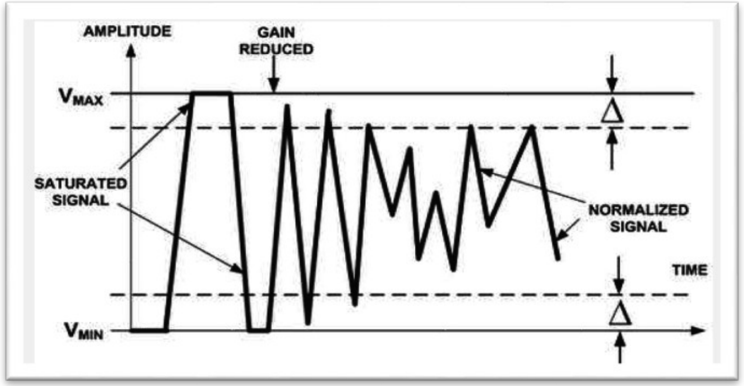
Experiment in the Lab
Dedicated software

- Used for acquiring signals (in time and frequency domain) of hammer load cell and 5 accelerometers.
- 12 hits per input to average, obtaining: **4x5 system** (collected in folders).
- Parameters to set up: sampling frequency, acquisition time.   ADREA RADICETTING

It allowed us to **discard invalid measurements**, such as:



Double hits

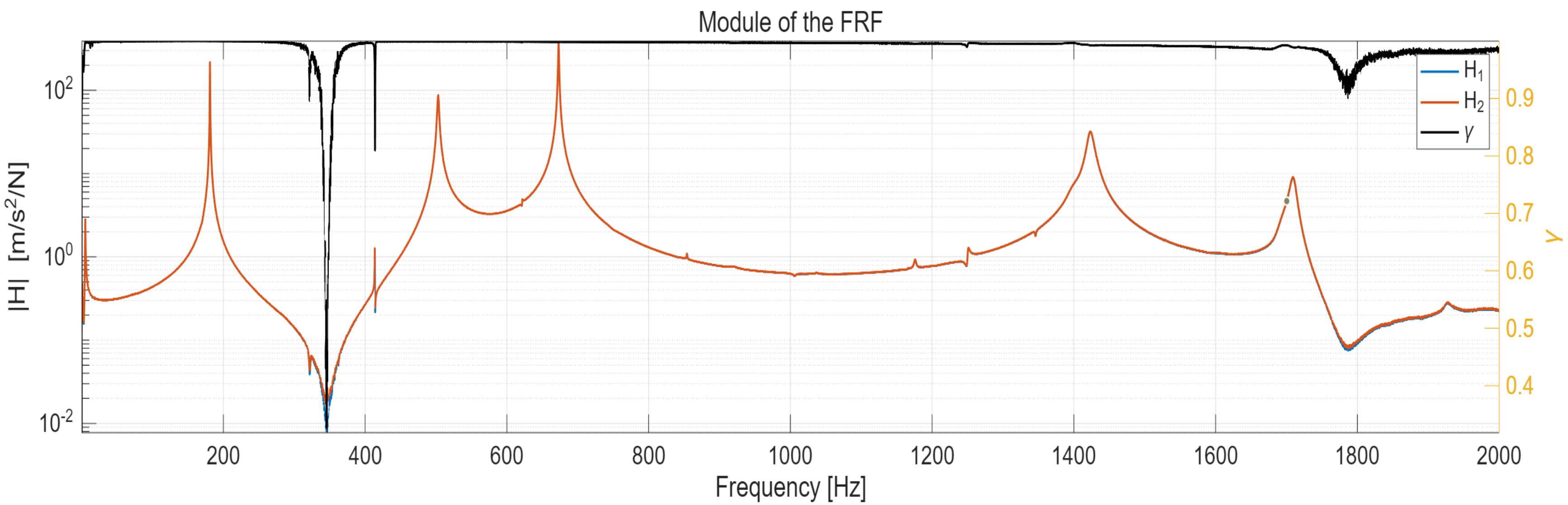


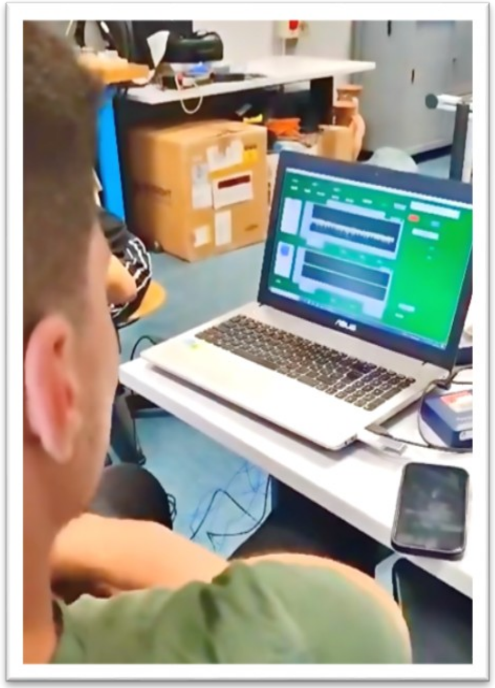
Too strong hits

Experiment in the Lab
Workflow approach

We adopted an **iterative validation process**, checking the consistency and correctness of the results at every step.

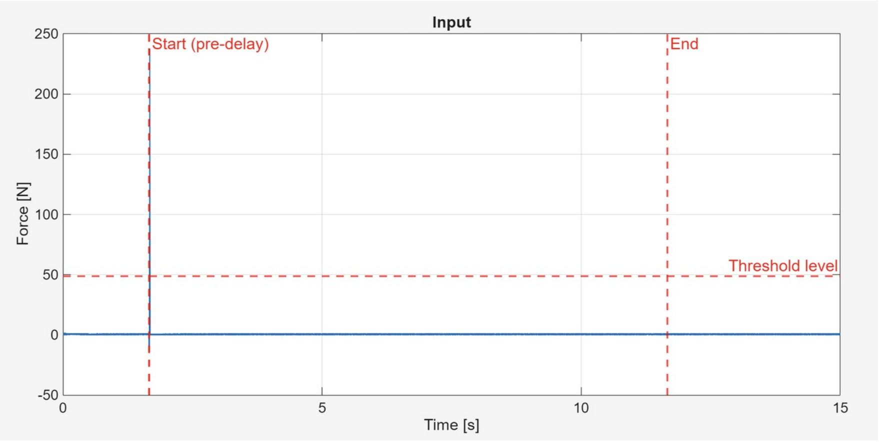
- After each test, the corresponding FRF assured us that the excited modes were the ones targeted by us.



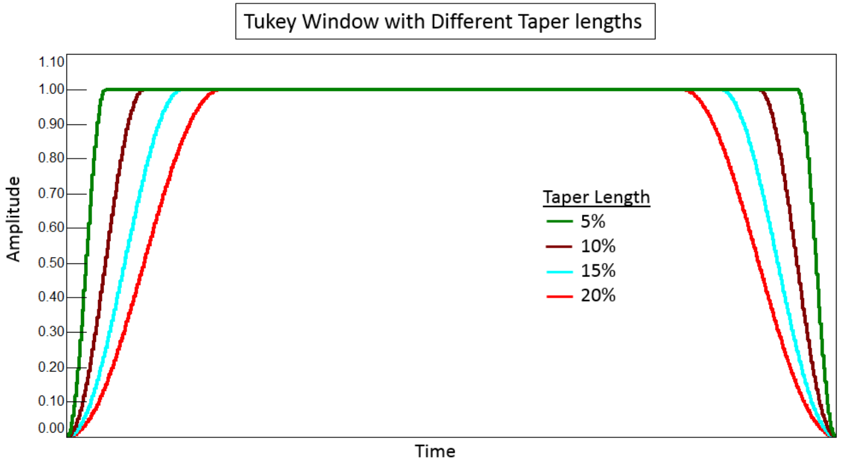


Post Processing - Cutting and Windowing

- The use of a **trigger** allowed us to start the acquisition when the trigger signal (input) exceeded a given **threshold**.
- The pre-trigger was set at 0.005s, defined by visual analysis.
- These techniques allowed for spectral averaging



- Double hits and strong hits were discarded manually not to contaminate the data.
- 50% **Tukey window** - General use, good compromise between leakage and resolution



Post Processing - Extraction of FRF's from Experimental Data

- Due to experimental phenomena, the generic transfer functions are unreliable, estimators are necessary.
- After extrapolation from the spectral data, both H1 and H2 were evaluated and plotted.
- The estimator chosen to represent the FRFs was **H1**.

Pros:

- Robust against noise on the output
- Good to throw off uncorrelated unmeasured inputs

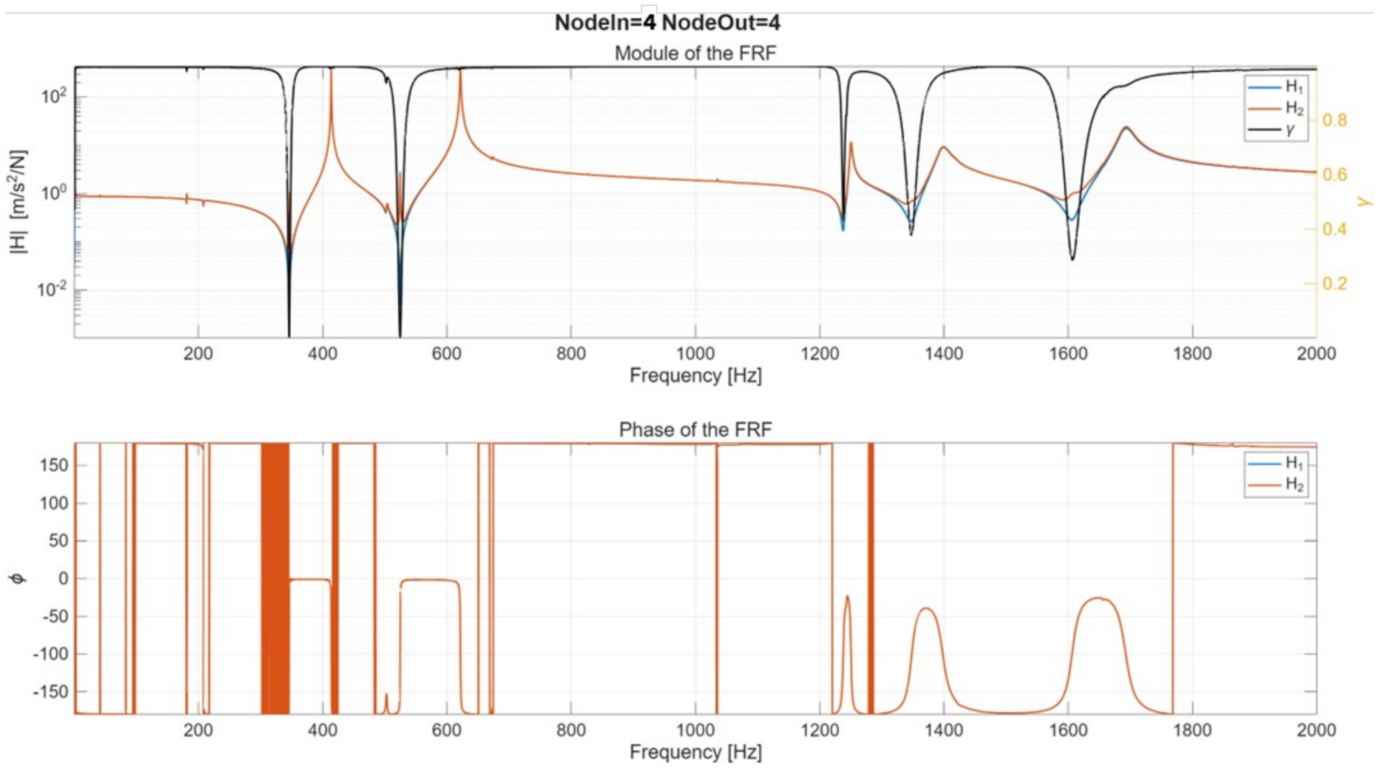
Cons:

- Susceptible against measured input noise
- Influenced by leakage

Solutions:

- Data cleaning, appropriate windowing, long acquisition time

Post Processing - Extraction of FRF’s from Experimental Data



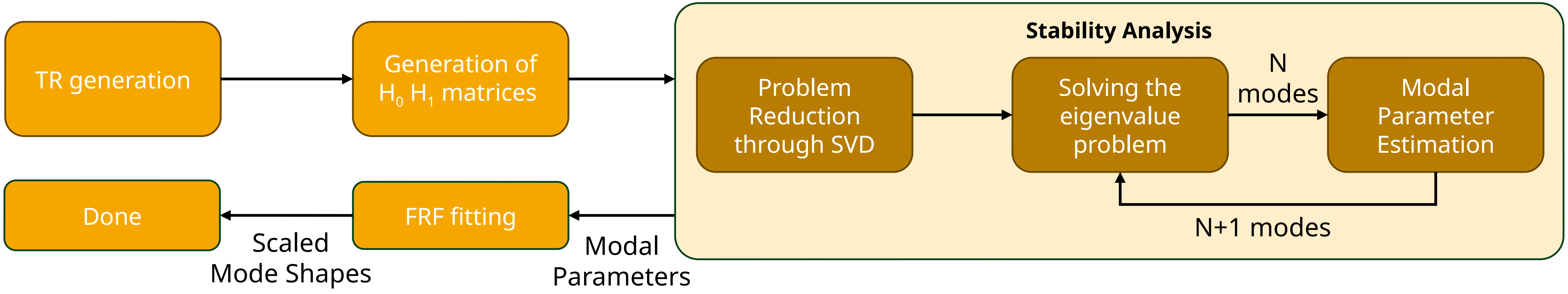
Immediate considerations

- Expected presence of anti-resonances in collocated TF
- Rough first visual estimates of the predicted eigenfrequencies and FRF behavior

Modal Identification
MDOF Identification with Ibrahim

- The system being identified is a **MIMO** system, with inputs being applied sequentially.
- Impact locations, and post processing of impulse data allows generation of input-output free time responses from the computed FRF's via the IFR.
- The generated time response data is directly usable in **Ibrahim** Algorithm.
- The FRF's show close peaks making SDOF identification unreasonable.

Algorithm Workflow

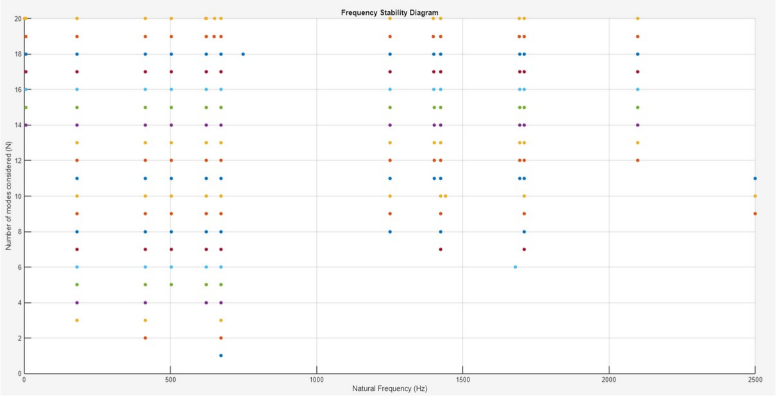


Modal Identification
MDOF Identification with Ibrahim – Parameter Selection

User defined parameters of the algorithm are;

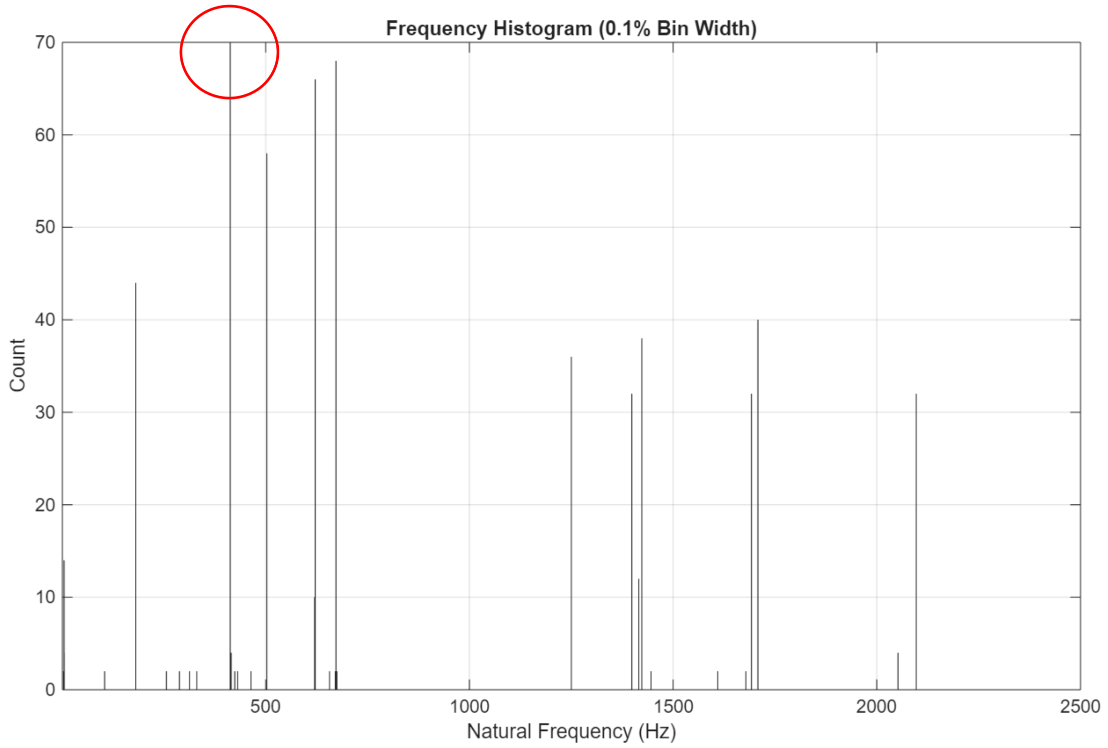
- **m and n** values, which define the maximum slicing of the time response data. Selection greatly influences the memory used by the algorithm and the length of time response used to compute modal parameters.
- The order of the problem is 2N, where N defines final count of complex conjugate pairs of modal parameters the algorithm outputs.

$$[H_{mn}(t)] = \begin{bmatrix} h(t) & h(t+\Delta t) & \dots & h\left(t+(n-1)\Delta t\right) \\ h(t+\Delta t) & h(t+2\Delta t) & \dots & h(t+n\Delta t) \\ \vdots & & & \\ h\left(t+(m-1)\Delta t\right) & h(t+m\Delta t) & \dots & h\left(t+(m+n-2)\Delta t\right) \end{bmatrix}$$



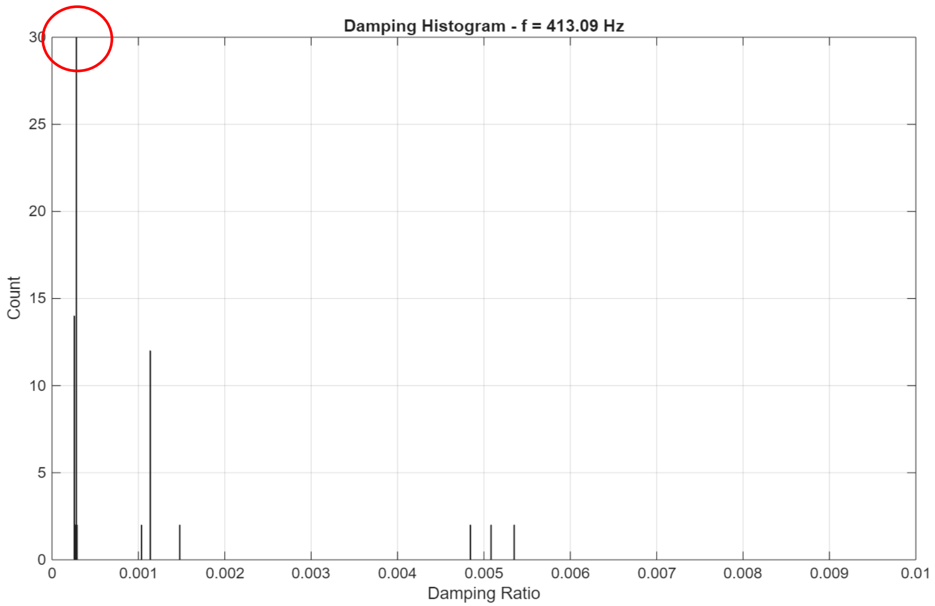
Modal Identification
Stability Analysis

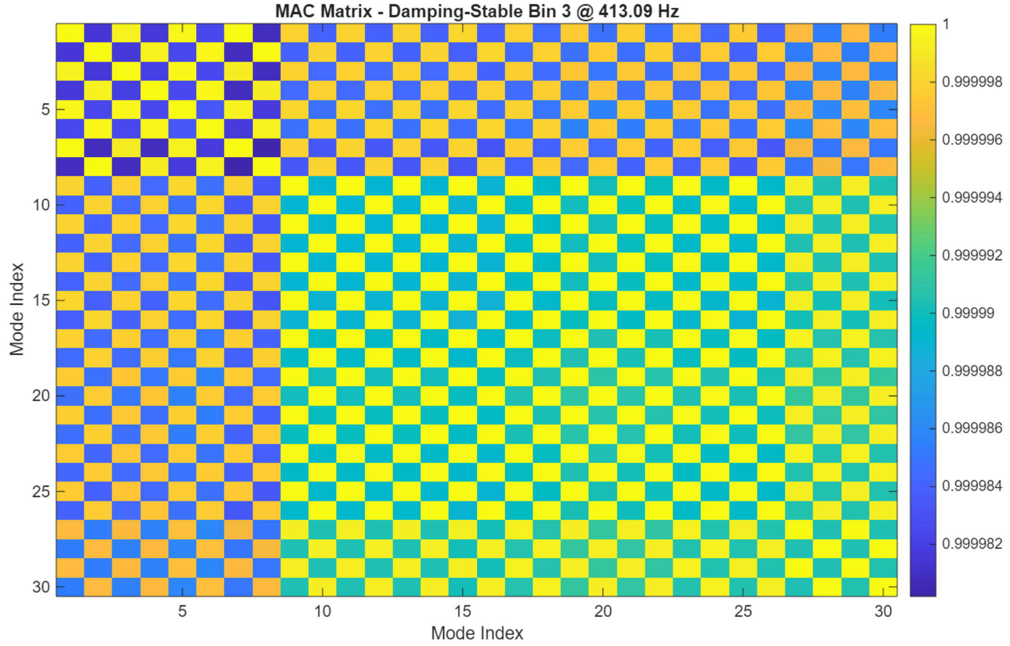
- The stability analysis of our approach was done in a sequential manner, with the subset of stable poles being subject to frequency, damping and mode shape stability analysis.
- For each obtained bin in the frequency stability histogram (+-0.1%), the most prominent ones were chosen as stable.



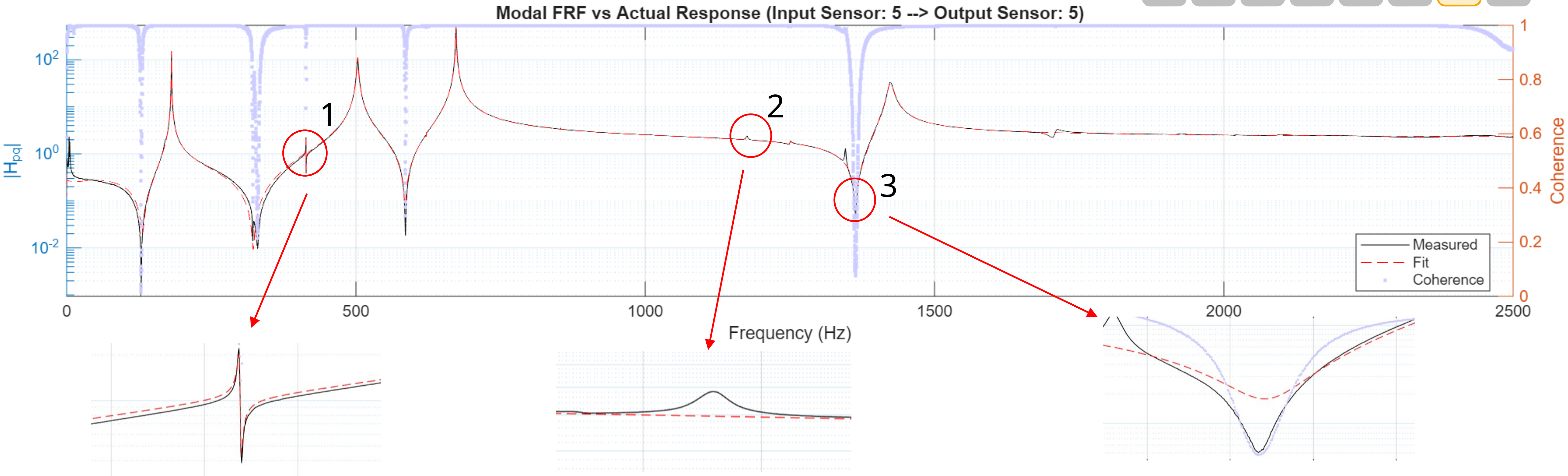
Modal Identification
Stability Analysis

- This procedure was repeated again for damping (+-5%).
- Finally, mode shape stability of both frequency and damping stable poles was analyzed through MAC.
- All the poles that proved stable in all domains were averaged in their characteristics and the resulting values were taken as reference.





Results, Validation, and Comparison

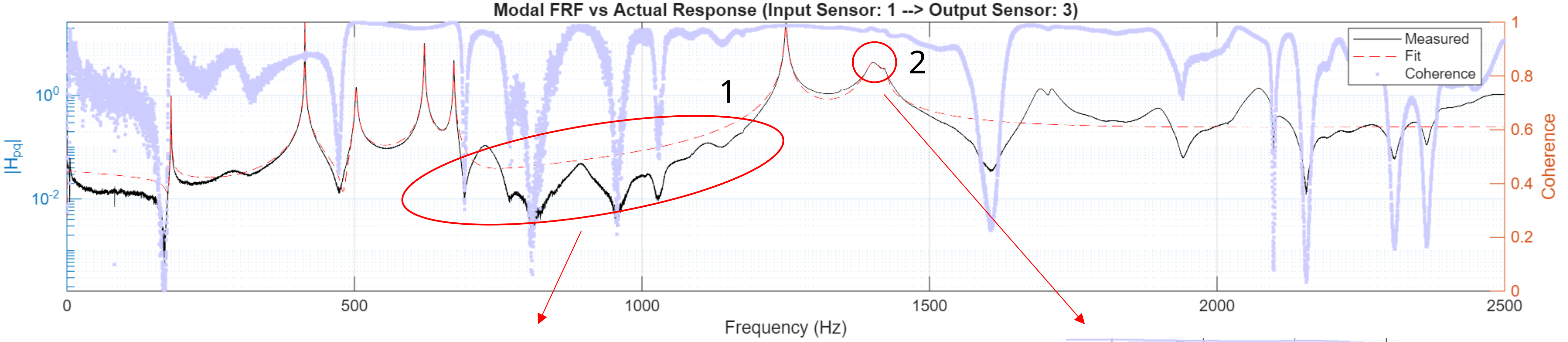


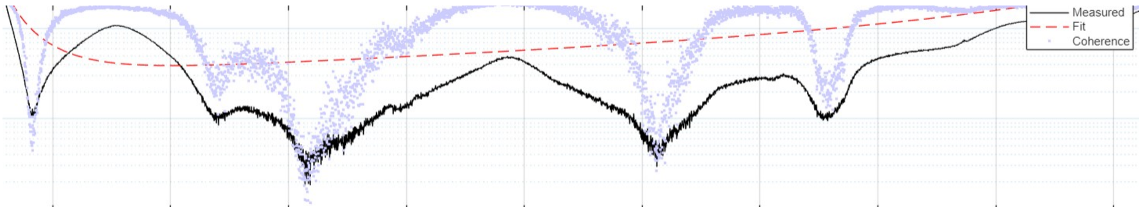
This corresponds to a **poorly excited mode**. The input-output pair (Input 5 → Output 5) **does not have strong participation** in that particular mode.

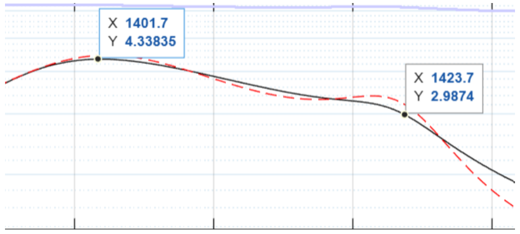
Small fluctuation in the experimental FRF is likely due to **uncorrelated input** at that frequency. The Ibrahim method typically does not reproduce this because it focuses on globally correlated modal dynamics.

Anti - resonance that lies in a low - energy region, that can be **dominated by noise**. However, **Ibrahim method emphasizes global modal behavior**, so the reconstructed FRF may smooth over the dip.

Results, Validation, and Comparison







The sharp dips seen in the experimental FRF are not captured by the Fit reconstruction, as they are not attributed to true system modes. These features **likely result from noise on the input** and Ibrahim focuses on globally correlated modal trends.

Ibrahim method **successfully identifies two closely spaced modes** around 1400–1425 Hz, demonstrating its **advantage over SDOF** approaches.

Comparison between experimental and numerical modal parameters

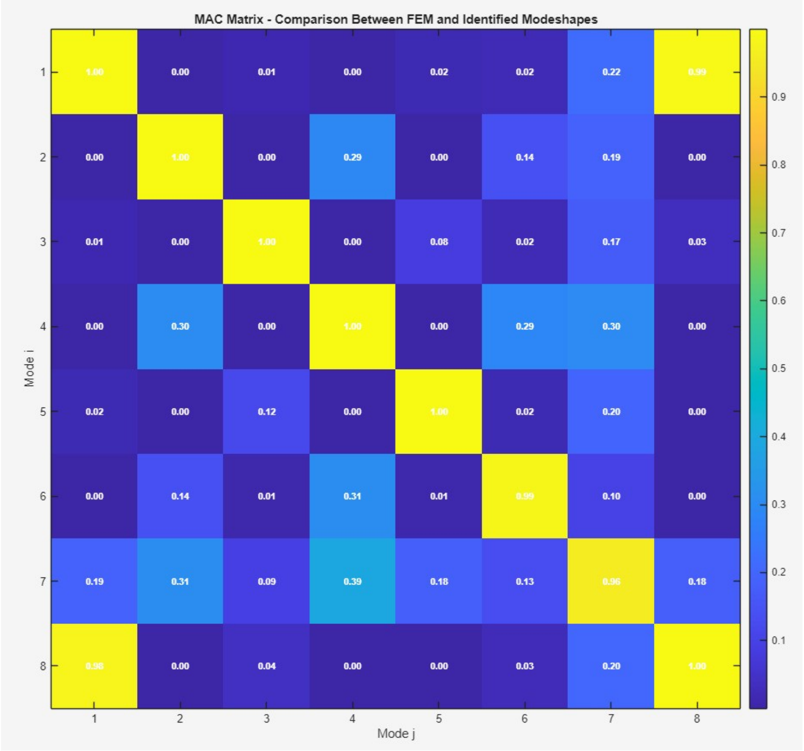
- **Eigenfrequencies** were directly compared in a summary table, showing good agreement with small differences due to simplifications in the FEM (e.g., idealized boundary conditions, no damping).

Mode	Measured (Hz)	Fit (Hz)	Damping Ratio ζ
1	178.68	181.13	0.0008
2	402.36	413.74	0.0003
3	485.17	503.02	0.0023
4	605.61	621.67	0.0007
5	654.59	672.98	0.0007
6	1233.9	1250.3	0.0011
7	1341.3	1398.8	0.0075
8	1383.3	1423.5	0.0032

Comparing mode shapes: Modal Assurance Criterion (MAC)

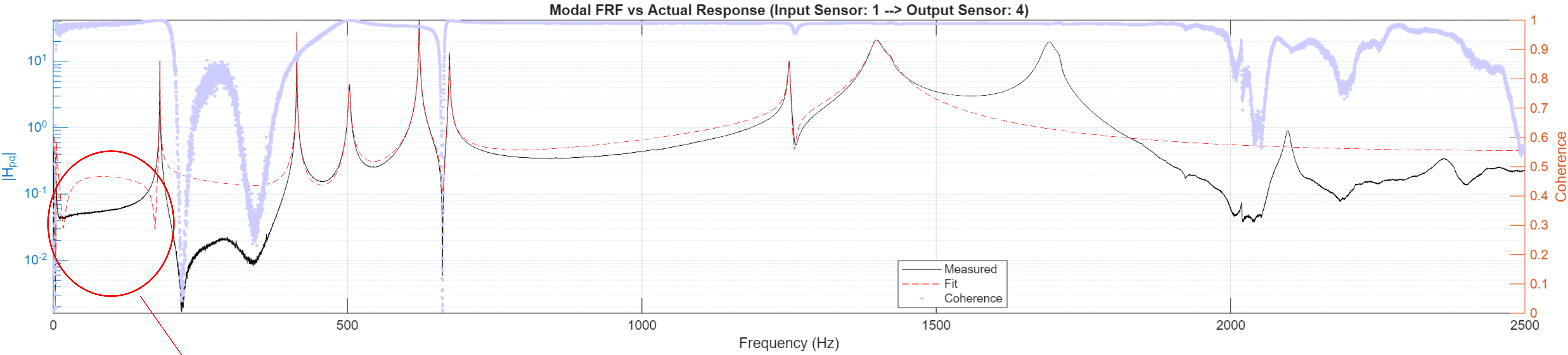
Given two mode shape vectors, the **MAC** is computed using the following formula:

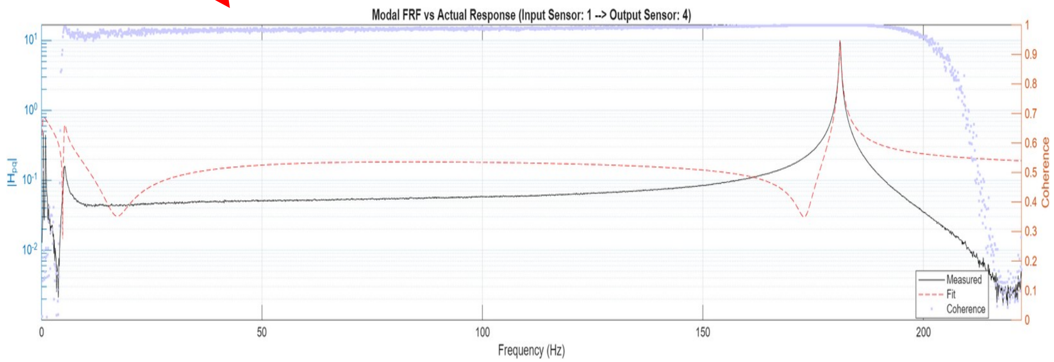
$$MAC_{(i,j)} = \frac{|\{\psi_i\}^T\{\psi_j\}|^2}{\left(\{\psi_i\}^T\{\psi_i\}\right)\left(\{\psi_j\}^T\{\psi_j\}\right)},$$



- The MAC matrix compares each pair of mode shapes from different datasets.
- Where i,j are respectively **numerical** and **experimental**.
- Useful for **validating** the consistency of extracted mode shapes.
- In our case it confirmed a **good match** with the FEM results.

Shortcomings





The static modes have not been fully identified. So coupling between low frequency modes and the dynamic ones cannot be captured with the modal FRF.

Conclusion ANDREA PUPPA

- We successfully carried out an **experimental modal analysis** on a T-shaped structure.
- The extracted modal parameters were in **good agreement with the FEM predictions**.
- The project demonstrated the effectiveness of our workflow, from modeling to post-processing.

Key takeaways:

- Importance of **proper sensor and input placement**
- Development of **custom post-processing tools**:
  we implemented dedicated MATLAB codes to compute **FRFs**, apply windowing, handle averaging and filtering, and prepare the data for reliable modal parameter extraction.
- Choosing the **appropriate modal identification method** (e.g., Ibrahim for MDOF)

Thank you